Investigation of single pool model of calcium

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Abstract

This paper serves as an investigation of the efficacy of the single pool calcium model suggested by Atri et al[1]. This model effectively describes intracellular calcium oscillations and waves in Xenopus laevis oocytes. It is comprised of a single IP_3 sensitive intracellular calcium pool which varies in size due to a negative feedback loop. The parameter μ, describing IP_3 receptor activation, was found to produce oscillatory behaviour within the range ($0.578 < \mu < 0.988$). When computed the model successfully reproduces experimental observations including calcium oscillations at constant IP_3 levels, wave propagation patterns, and response characteristics to varying stimulation levels that result in stable transient mechanics. The model demonstrates how calcium signalling transitions from steady-state to oscillatory behaviour through Hopf bifurcations, with oscillation frequency and amplitude increasing as μ increases.

1 Introduction

Calcium is crucial for cell dynamics and regulation. Consequently, intracellular calcium concentration is tightly regulated due to its essential role in signal transduction pathways [2], its function as a secondary messenger in neurotransmitter release, and its involvement in maintaining the membrane potential. Due to the importance of calcium regulation, many models have been developed to detail these intricate processes. The paper "A Single-Pool Model for Intracellular Calcium Oscillations and Waves in the Xenopus laevis Oocyte" by Atri et al is one such attempt at simplifying these complex dynamics [1]. In order to counter the complexity, the paper presents an abstracted mathematical model for calcium oscillations, which attempts to capture the essential mechanisms responsible for the complex spatiotemporal patterns of intracellular calcium in these cells. This system is unique to the Xenopus oocyte- a large egg from the african clawed frog. This cell is of interest due to the low background of endogenous channels and receptors, as well as its ability to be maintained in vitro for extended periods, allowing for experiments to be performed. This allows for large amounts of experimental data to be acquired and to subsequently build a model from the observational data. The model developed is based on a single intracellular calcium pool and the properties of the inositol 1,4,5-trisphosphate (IP_3) receptor, which serves as a calcium channel.

2 Explanation of Calcium cycle

There are four key stages of calcium homeostasis in the Xenopus oocyte: initialisation, amplification, inhibition, and recovery. This cycle repeats periodically and is reflected in the model through the oscillatory behavior of key parameters. Specifically, **c** represents the intracellular calcium concentration (measured in micromolar), while **n** is a dimensionless variable indicating the proportion of unclosed IP_3R receptors on the endoplasmic reticulum, which regulate calcium release. These parameters are closely linked: IP_3 receptors are inhibited by calcium binding, but in the absence of calcium, the receptors remain open, allowing continuous release of stored Ca^{2+}.

The cycle starts when an external stimulus/ agonist binds to a receptor that triggers the production of IP_3 within the cell. This IP_3 diffuses through the cytoplasm and binds to IP_3 receptors ($IP_3R's$) in the endoplasmic reticulum (ER), causing them to open and release stored calcium. In the model this mechanism, which details the rate of change of the calcium pool in the cell is described by Eq:1

$$\frac{dc}{dt} = J_{channel} - J_{pump} + J_{leak} \tag{1}$$

This equation represents the rate of change of intracellular calcium concentration (**c**) over time. It is governed by three key components: $J_{channel}$, which represents the influx of calcium through calcium channels;J_{pump}, the active removal of calcium from the cell by calcium pumps; and J_{leak}, a passive, continuous calcium intake to the cell due to the membrane permeability. $J_{channel}$ and J_{leak} are defined in equations [2,3]

$$J_{channel} = k_{flux}\mu([IP_3])n(b + \frac{V_1 c}{k_1 + c} \tag{2}$$

$$J_{leak} = \beta \tag{3}$$

The released calcium activates nearby $IP_3R's$, creating a positive feedback loop that amplifies the calcium release. This causes the concentration to increase locally and spread as a wave throughout the cell[3]. The amplification is represented in the model by the term $\frac{V_1 c}{k_1+c}$ in equation 2.

Additionally μ, a function of IP_3, serves as a key parameter in the model that links the concentration of IP_3 to the activation state of IP_3R's, thereby influencing calcium signalling dynamics. μ controls the strength of the calcium signalling response, with higher values leading to more intense and complex oscillatory behaviour. This represents how strongly the IP_3 receptors are stimulated, which in turn affects the calcium release from internal stores in the ER and the overarching calcium dynamics in the cell. Similarly to **n**, the value μ is dimensionless.

However, as the calcium levels rise, the $IP_3R's$ are inhibited, subsequently slowing the release of calcium release. This inhibition is described by the equation 4, which governs the rate of change of unclosed $IP_3R's$:

$$\tau_n \frac{dn}{dt} = n_\infty(c) - n \tag{4}$$

Where:

$$n_\infty(c) = 1 - \frac{c^2}{k_2^2 + c^2} \tag{5}$$

With this inhibition the cell is able to return to its starting state as the Calcium pumps on the ER and the cell membrane remove calcium from the cytoplasm back into the ER and out of the cell. This process is represented by the J_{pump} term in equation 6:

$$J_{pump} = \frac{\gamma c}{k_\gamma + c} \tag{6}$$

As calcium levels decrease, the inhibition of $IP_3R's$ is reduced, returning the cell to its starting position allowing the cycle to repeat once an agonist binds to the external receptor.

The spatial aspect of calcium dynamics is not explicitly represented in these equations, which describe the temporal evolution of calcium concentration at a single point. However, the spatial propagation of calcium waves can be modeled by extending these equations to include diffusion terms and considering the spatial distribution of IP_3 and calcium.

These equations can be combined to form two key systems equations that describe the dynamics of cytosolic calcium concentration (**c**) and the proportion of $IP_3R's$ not closed by calcium (**n**):

$$\frac{dc}{dt} = k_{flux}\mu([IP_3])n(b + \frac{V_1 c}{k_1 + c}) - \frac{\gamma c}{k_\gamma + c} + \beta \tag{7}$$

$$\tau_n \frac{dn}{dt} = 1 - \frac{c^2}{k_2^2 + c^2} - n \tag{8}$$

3 Model Simulation against time

To assess the effectiveness of the model, simulations were conducted over a time span of 50 time units(seconds), ensuring a minimum of three full oscillatory cycles. The model variables were plotted as functions of time under three different initial conditions and three distinct values of the parameter μ, representing IP_3. The model was implemented into MATLAB(shown in the code library). The oscillatory range was determined by identifying the Hopf bifurcation points of μ in MATCONT. This range was found to be $0.57799236 < \mu < 0.98999332$, closely aligning with the values reported in the original study by Atri[1].

The presence of two Hopf points confirms that the system undergoes a supercritical Hopf bifurcation as μ increases, leading to the emergence of stable limit cycles (oscillations) within this range. Physiologically, this range has significant implications. For $\mu < 0.57799236$, the system will remains in a stable low-calcium state. As μ surpasses this threshold, oscillations emerge, reflecting the feedback cycle associated with calcium dynamics (explained in the calcium model section). The relatively broad oscillatory range ($\Delta\mu$=0.412)

Plots of calcium content of the cell and proportion of IP_3 full over time

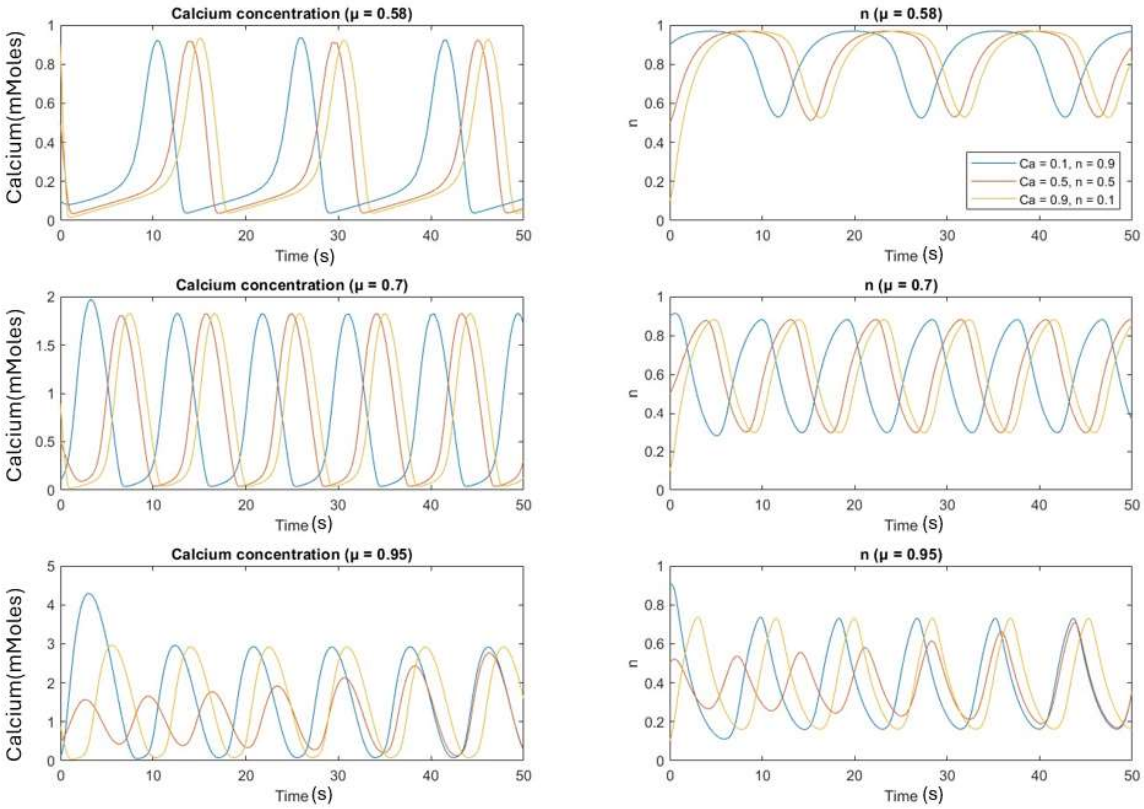

Figure 1: Plot detailing the calcium concentration of the cell and the proportion of IP_3 receptors bound by calcium for different values of μ, for three different sets of initial conditions

suggests that the system can sustain calcium signalling across varying IP_3 concentrations. Beyond $\mu =$ 0.990, the system transitions to a stable high-calcium state, indicating a fixed region of oscillatory behaviour.

Firstly, when μ =0.58-a value close to that of the first Hopf Bifuracation- it can be observed that the amplitude of the calcium oscillations is lessened, falling between 0 and 1 mMoles and displays a regular periodic behaviour. The frequency of the waves is lower, with only three complete cycles over 50 seconds. It can also be seen that the three initial conditions quickly converge to the same periodic pattern, and that the IP_3 channel dynamics, represented by **n**, oscillate in a tight region between 0.5-0.99. This is common across all initial conditions.

In the second row of graphs $\mu = 0.7$ which is a near midpoint of the oscillatory range. This higher value of μ results in an increased amplitude (0-2 mMoles) of the calcium waves, in addition to a higher frequency (5 cycles/50 seconds), with more pronounced peaks. Similarly to $\mu = 0.58$ the waves quickly converge to stable oscillations. The dynamics of the IP_3 receptors has several pronounced differences when compared to the first case: firstly, there is a larger oscillation range (0.3-0.9); more rapid transitions and most notably a near antiphasal relationship with calcium- when n is maximum, the calcium takes its minimum.

Finally, at $\mu = 0.95$(Near Second Hopf Point), the calcium oscillations show yet larger oscillations, going to a max of 4.3 mMoles. However, this is only the case for the first spike of the low calcium initial condition, as the wave is dampened after the first oscillation and remains stable at 3.0. This contrasts with the middle condition(Ca=0.5mMoles, **n**= 0.5) which has an initial small spike of 1.6mMoles that builds to the max amplitude of 3.0mMoles over the course of the period. Despite the initial complex transients, by the end of the time period all conditions are near in-phase.

An examination of Fig:1, brings to light many key behaviours of the model. It can be seen that as μ increases, the system becomes more complex, with ever-increasing amplitude and frequency; the initial transients become more pronounced in the difference between the amplitude between the initial and final peak of the period.

However, it can be seen in all cases that eventually a stable periodic behaviour emerges and a clear

Phase Plane plot of Initial conditions for different values of μ

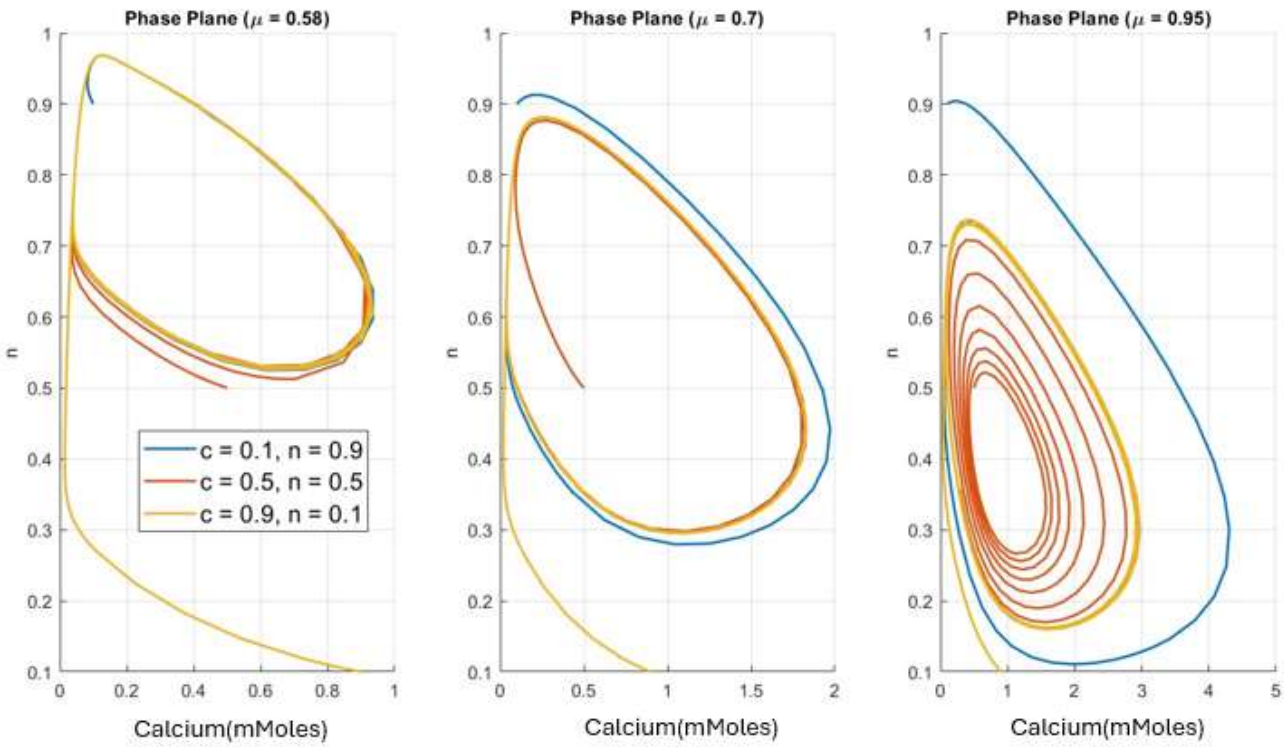

Figure 2: Represents the phase planes of three different initial conditions-variation in calcium and proportion of IP_3-where each plane corresponds to a different value of μ

coupling between calcium and IP_3 dynamics. Additionally, the final state of the system is irrespective of the initial conditions of the system.

4 Dynamical Systems Analysis

4.1 Phase Plane Investigation

These relations can be demonstrated more clearly when investigating the trajectories of the wave in the phase plane. The model was again simulated over a time period of 50 seconds for three different conditions of high and low proportions of calcium and proportion of IP_3 closed.

The values again were chosen as they suitably demonstrate the dynamical range of the model, with the values of μ being between the two Hopf points.

As is shown in the Fig:2 the behaviour of the model across the oscillatory range of μ varies greatly. As in Fig:1, the model is assessed across three different initial conditions and three distinct values of μ.

In the first plot μ takes the value of 0.58, which is located near the first Hopf Bifurcation. This plot illustrates a small-amplitude limit cycle with $[Ca^{2+}]$ confined to a range of [0, 1]mMoles, this is reflected previously in Fig:1. Additionally, the trajectories converge relatively quickly to the limit cycle. The variable n also only oscillates in a narrow range (0.5-0.9). This behaviour is typical of a system near a Hopf bifurcation.

At the intermediate point of $\mu = 0.7$, the system exhibits an increased amplitude of the Limit Cycle. This results in increases to both c and n, with c reaching 2mMoles and **n** a range of [0.28,0.9]. More pronounced periodic behaviour with consistent convergence patterns. However, there is a more pronounced difference between the initial conditions with them spreading more within the cycle and being more discernable.

Finally at $\mu = 0.95$ (near the second Hopf Bifuracation), there is an even larger limit cycle which shows the calcium concentration reaches 5 units high. Additionally, the even split calcium and IP_3 initial conditions, represented in red form a complex spiral trajectory. It is shown that multiple rotations were

Bifuracation diagram of model showing oscillatory region and limits

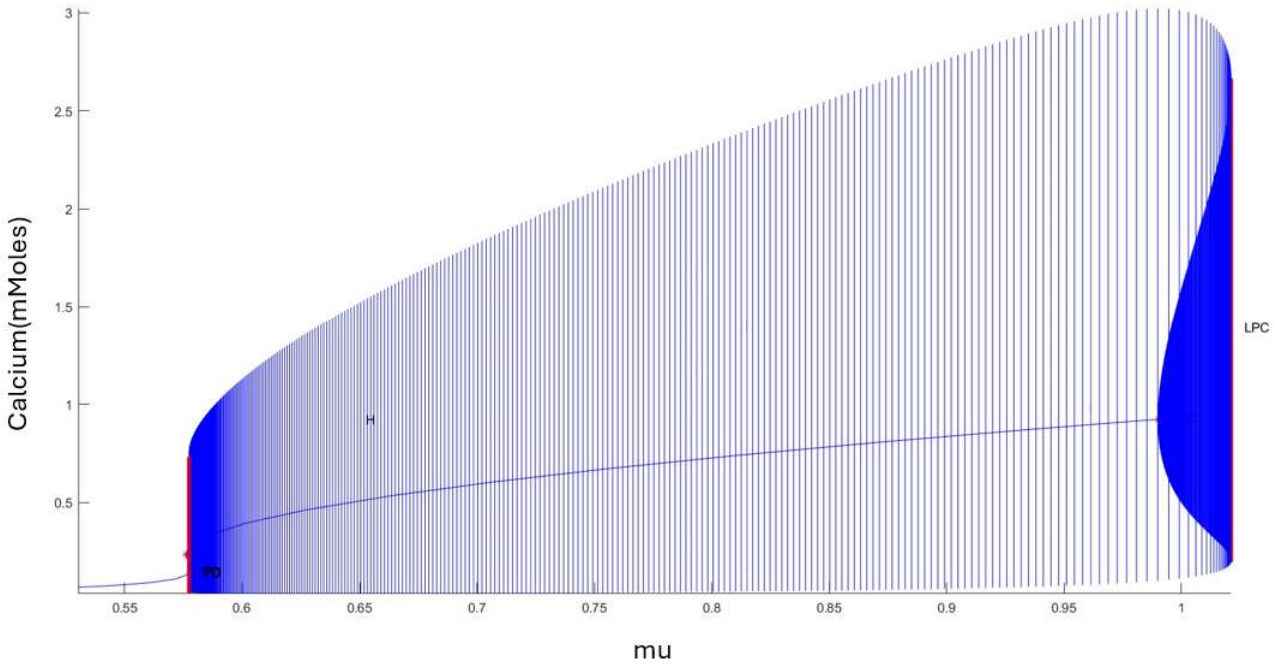

Figure 3: Bifurcation diagram of the second Hopf point, with calcium content on the Ordinate axis, and bifurcation parameter μ along the abscissa.

required before reaching a stable pereodic orbit. This inconsistency is also highlighted in Fig:1 where the second IC results in slowly increasing amplitude of calcium.

These individual dynamics demonstrate 3 key behaviours of the model. Firstly, the evolution of the amplitude as μ increases. There is an increase in oscillation amplitude with increasing μ that results in both an expansion of phase space coverage via a wider range of both calcium concentration and n-variable.

Secondly, the model displays convergent properties for all values of μ within the oscillatory range, where stable limit cycles are formed. However, longer convergence times are observed for $\mu = 0.95$. This means that the transient mechanics- the path by which the model returns to an equilibria- are more complex.

Finally, the system displays strong stability, in that it will return to stable oscillations, allowing for highly predictable long-term behaviour irrespective of initial conditions. These trends clearly demonstrate that this model is very suitable for representing calcium homeostasis. The system will quickly return to stable oscillations.

4.2 Bifurcation Analysis

To greater understand the system a Bifurcation diagram can be used to illustrate the dynamics. Bifurcation's provide critical insights into important parameter values which cause behaviour changes, such as Hopf and limit Points, and how the amplitude of the oscillations vary with these changes. Additionally, it can serve to identify stable and unstable solutions and show where qualitative changes in dynamics occur. Whilst the previous phase planes were snapshots of the dynamics at specific value of μ Fig:3, illustrates the complete range of calcium concentrations (0.2 to 3.0 units) and clearly shows the suitability of the model to maintain controlled oscillations.

As is shown in Fig:3, there is a clear transition from stable equilibria to oscillatory behaviour across the range of μ values. This oscillatory range is represented by the vertical lines which represent individual limit cycle amplitudes. These lines show a progressive increase in oscillation amplitude from $\mu > 0.57$ to $\mu < 1$. It is shown that the maximum calcium concentration reaches approximately 3.0 units. Whilst, this may seem counter to Fig:2, where the maximum amplitude of calcium oscillation was 4.3 units, the bifurcation instead represents the stable oscillations of the system. The minimum value of calcium under stable oscillations is 0.2

units. Near the Hopf points it can be seen that there are dense vertical lines, which indicate high-frequency oscillations. This is supported by Fig1, where at the upper limit there is great frequency, however, this is not reflected for the low μ value.

Fig:3, contains three key regions: a non-oscillatory region before the first Hopf point, where a single stable equilibrium branch exists. This is characterised by a low calcium, content and no oscillations. After this a period doubling point is found, this indicates the transition where the system shifts from steady-state behaviour to oscillatory dynamics, indicating where the period of oscillation doubles and new solution branches are found. Within the oscillatory region it can be observed that stable limit cycles emerge, with gradually increasing amplitude and a continuous growth in mean calcium level, where a regular spacing of vertical lines suggests a consistent period. After this period ($\mu > 0.998$) the terminal region begins which displays a rapid expansion of oscillation amplitude and complex dynamics near LPC point.

This bifurcation diagram effectively illustrates the transitions from steady-state to oscillatory behaviour and demonstrates how the IP_3 receptor activation (μ) controls the system's dynamic properties.

5 Conclusion

This model serves as an effective tool for understanding the complex interplay of factors regulating intracellular calcium dynamics. Within the oscillatory range of μ, it reliably re-establishes stable transient behaviour regardless of initial conditions, making it well-suited for mimicking homeostatic processes.

The model's robustness is demonstrated by its consistent convergence to stable limit cycles, as shown by its predictable transitions between steady states and the phase-plane analysis in Fig:2, which highlights the coupling between calcium concentration and IP_3 receptor states. As μ increases, the model exhibits larger calcium wave amplitudes and more complex dynamics, reflecting parameter-dependent behaviour captured in the bifurcation diagram in Fig:3.

These features make the model a highly accurate representation of calcium homeostasis mechanisms, showing reliable reproduction of experimental observations. Its ability to consistently return to stable oscillations makes it a powerful tool for studying calcium signalling in Xenopus laevis oocytes, offering valuable insight into cellular regulation while maintaining mathematical simplicity.

References

[1] Alireza Atri et al. "A single-pool model for intracellular calcium oscillations and waves in the Xenopus laevis oocyte". In: *Biophysical Journal* 65.4 (1993), pp. 1727–1739.

[2] Laura Fedrizzi, Dmitry Lim, and Ernesto Carafoli. "Calcium and signal transduction". In: *Biochemistry and Molecular Biology Education* 36.3 (2008), pp. 175–180.

[3] H Llewelyn Roderick, Michael J Berridge, and Martin D Bootman. "Calcium-induced calcium release". In: *Current Biology* 13.11 (2003), R425.

6 Code Library

```
% Model parameters
k_flux = 8.1;
b = 0.111;
V1 = 0.889;
k1 = 0.7;
gamma = 2.0;
k_gamma = 0.1;
beta = 0.01;
tau_n = 2.0;
k2 = 0.7;

% Time span for integration
tspan = [0 50];

% Different initial conditions
initial_conditions = [0.1, 0.9; 0.5, 0.5; 0.9, 0.1];

% Different mu values
mu_values = [0.58, 0.7, 0.95];

%% Time Series Plots (Calcium & n vs Time)
figure('Name', 'Time Series Plots', 'Position', [100, 100, 1400, 800]);

for i = 1:length(mu_values)
    mu = mu_values(i);

    % Plot Calcium Concentration vs Time
    subplot(2, length(mu_values), i);
    hold on;

    for j = 1:size(initial_conditions, 1)
        y0 = initial_conditions(j, :);
        [t, y] = ode45(@(t, y) calcium_model(t, y, mu, k_flux, b, V1, k1, gamma, k_gamma, beta
            , tau_n, k2), tspan, y0);
        plot(t, y(:, 1), 'LineWidth', 1.5); % Calcium concentration
    end

    title(['[Ca^{2+}] vs Time (\mu = ', num2str(mu), ')']);
    xlabel('Time (s)');
    ylabel('[Ca^{2+}]');
    legend({'c = 0.1, n = 0.9', 'c = 0.5, n = 0.5', 'c = 0.9, n = 0.1'}, 'Location', 'best');
    grid on;

    % Plot n vs Time
    subplot(2, length(mu_values), length(mu_values) + i);
    hold on;

    for j = 1:size(initial_conditions, 1)
        y0 = initial_conditions(j, :);
        [t, y] = ode45(@(t, y) calcium_model(t, y, mu, k_flux, b, V1, k1, gamma, k_gamma, beta
            , tau_n, k2), tspan, y0);
        plot(t, y(:, 2), 'LineWidth', 1.5); % n variable
    end

    title(['n vs Time (\mu = ', num2str(mu), ')']);
    xlabel('Time (s)');
    ylabel('n');
    grid on;

    hold off;
end

%% Phase Plane Plots (Calcium vs n)
figure('Name', 'Phase Plane Plots', 'Position', [150, 150, 1200, 600]);
```

```

for i = 1:length(mu_values)
    mu = mu_values(i);

    subplot(1, length(mu_values), i);
    hold on;

    for j = 1:size(initial_conditions, 1)
        y0 = initial_conditions(j, :);
        [t, y] = ode45(@(t, y) calcium_model(t, y, mu, k_flux, b, V1, k1, gamma, k_gamma, beta
            , tau_n, k2), tspan, y0);
        plot(y(:, 1), y(:, 2), 'LineWidth', 1.5);
    end

    title(['Phase Plane (\mu = ', num2str(mu), ')']);
    xlabel('[Ca^{2+}]');
    ylabel('n');
    legend({'c = 0.1, n = 0.9', 'c = 0.5, n = 0.5', 'c = 0.9, n = 0.1'}, 'Location', 'best');
    grid on;

    hold off;
end

%% ODE System Function
function dydt = calcium_model(~, y, mu, k_flux, b, V1, k1, gamma, k_gamma, beta, tau_n, k2)
    c = y(1);
    n = y(2);

    dcdt = k_flux * mu * n * (b + (V1 * c) / (k1 + c)) - ...
           (gamma * c) / (k_gamma + c) + beta;
    dndt = (n_inf(c, k2) - n) / tau_n;

    dydt = [dcdt; dndt];
end

%% Steady-State Function for n
function n_inf_val = n_inf(c, k2)
    n_inf_val = 1 - (c^2) / (k2^2 + c^2);
end
```

Listing 1: MATLAB implementation of Calcium Waves model as described in the paper by Atri et al, generates graphs of c/n , t and phase plane plot of c,n for 3 different values of μ